

Conv2d as AvgPool2d

What is it 💡?

This function is used to **replace an AvgPool2d** operation with a **Conv2d** operation, which is beneficial for deploying on hardware, backpropagation control and compatibility.

To replace an **average pooling** operation (**AvgPool2d**) with a **convolutional layer** (**Conv2d**), we need to construct a **depth-wise convolution** that mimics the behavior of average pooling.

TL; DR: In this guide, we imitate AvgPool2d with Conv2d. Code: [📄 Conv2d as AvgPool2d | Weight](#)

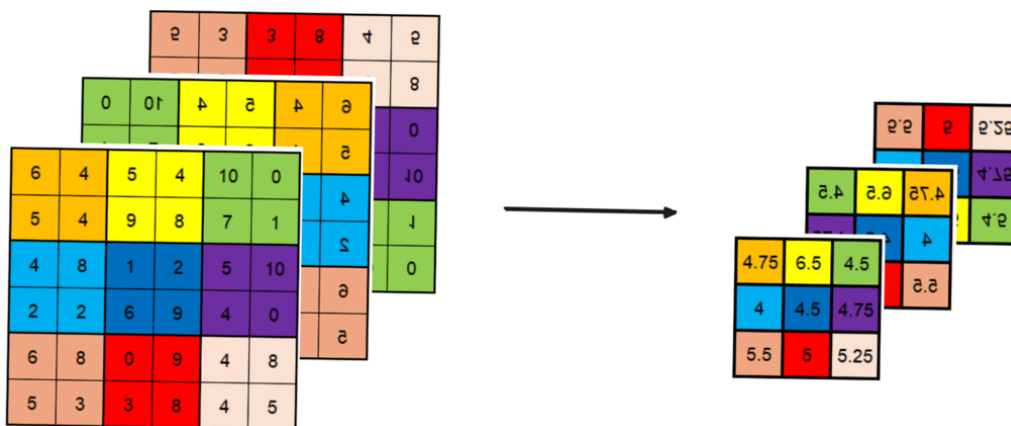
1.1 What Does Average Pooling Do?

Average pooling computes the mean of values in a local window (kernel) across spatial dimensions (height, width).

For a given kernel of size $[k, k]$, it replaces each region with:

$$\text{New Pixel} = \frac{1}{k \times k} \sum \text{Values in the window}$$

For example, a $[2, 2]$ average pooling with stride 2 operates like this:



⇒ To replace average pooling, we need a **depth-wise convolution layer** that applies a kernel with uniform weights.

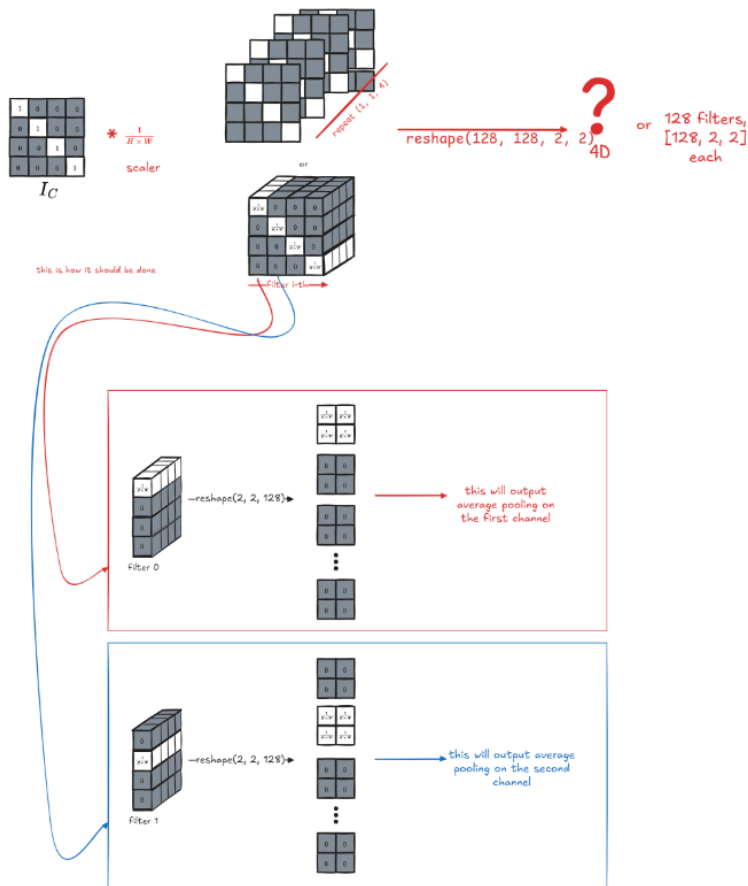

```

8     ).reshape(channels, channels, kernel_size[0], kernel_size[1])
9
10    parameters = { "weight": ttnn.from_torch(conv_weights, dtype=dtype) }
11
12    return parameters
13
14
15    """ Example usage"""
16    # avgpool2d_to_conv_parameters = get_conv2d_for_avgpool2d(
17    #     kernel_size=[2, 2],
18    #     channels=64,
19    #     dtype=ttnn.bfloat16,
20    # )
21
22    # ttnn_avg_pool = Conv(
23    #     [1, 1, 0, 0],
24    #     ttnn_tensor.shape,
25    #     parameters=avgpool2d_to_conv_parameters,
26    #     dtype=ttnn.bfloat8_b,
27    # )
28
29

```

Weight Visualization

The parameters for the convolution layer can be illustrated like this, I presumed:



Experimental Result

- Comparing to executing AvgPool2d on cpu - **22.9x faster** while still retaining a good PCC.

	Device	Time (μ s)	$\pm\mu$ s	PCC
BatchNorm2d				
Fallback	CPU	262.0	14.3	0.999966
Workaround	Wormhole	97.3	2.19	
avgpool2d				
Fallback (CPU)	CPU	2610.0	30.8	0.999812
Workaround	Wormhole	114.0	22.1	